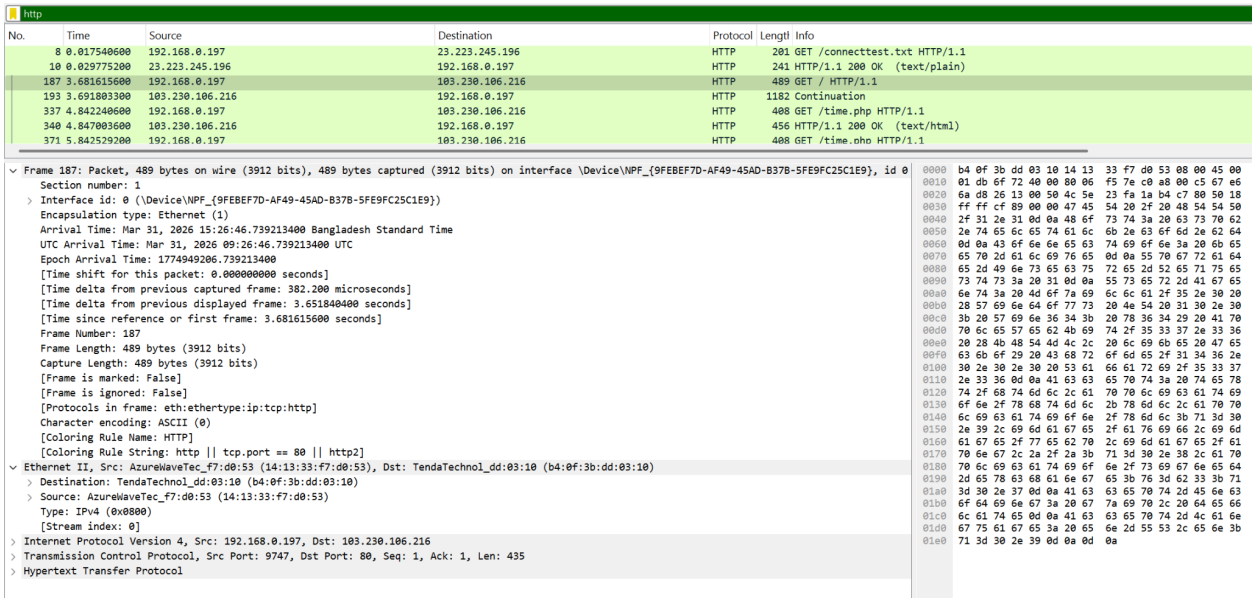
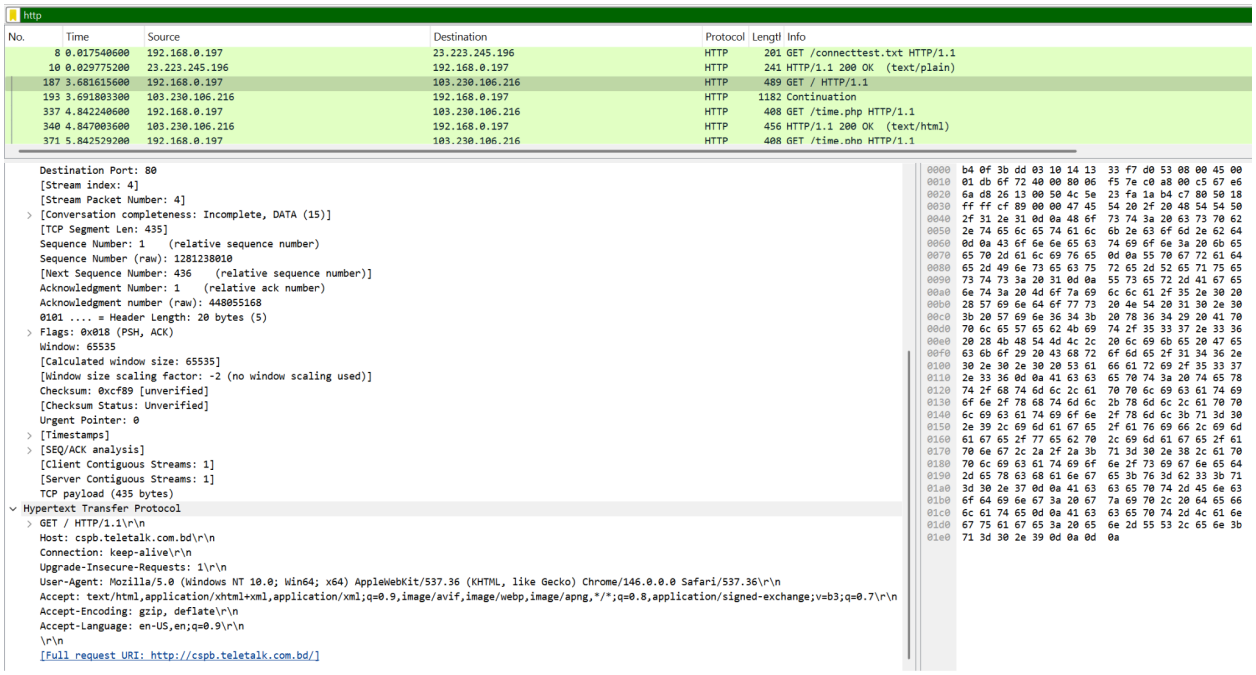
HTTP Request Packet:



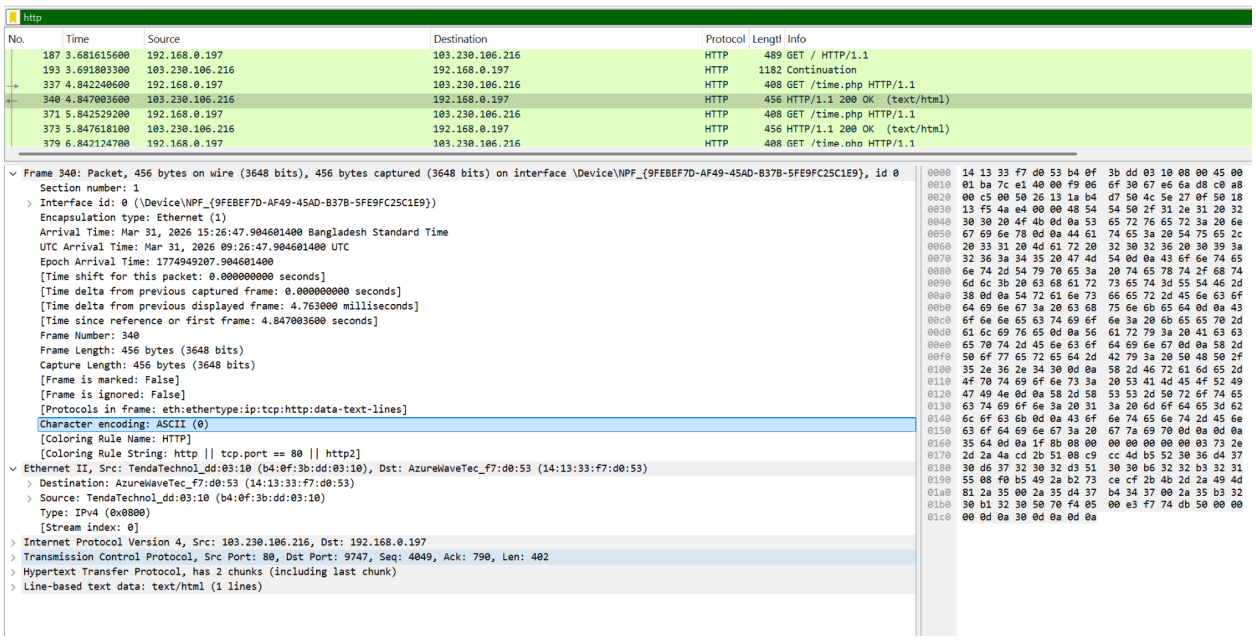


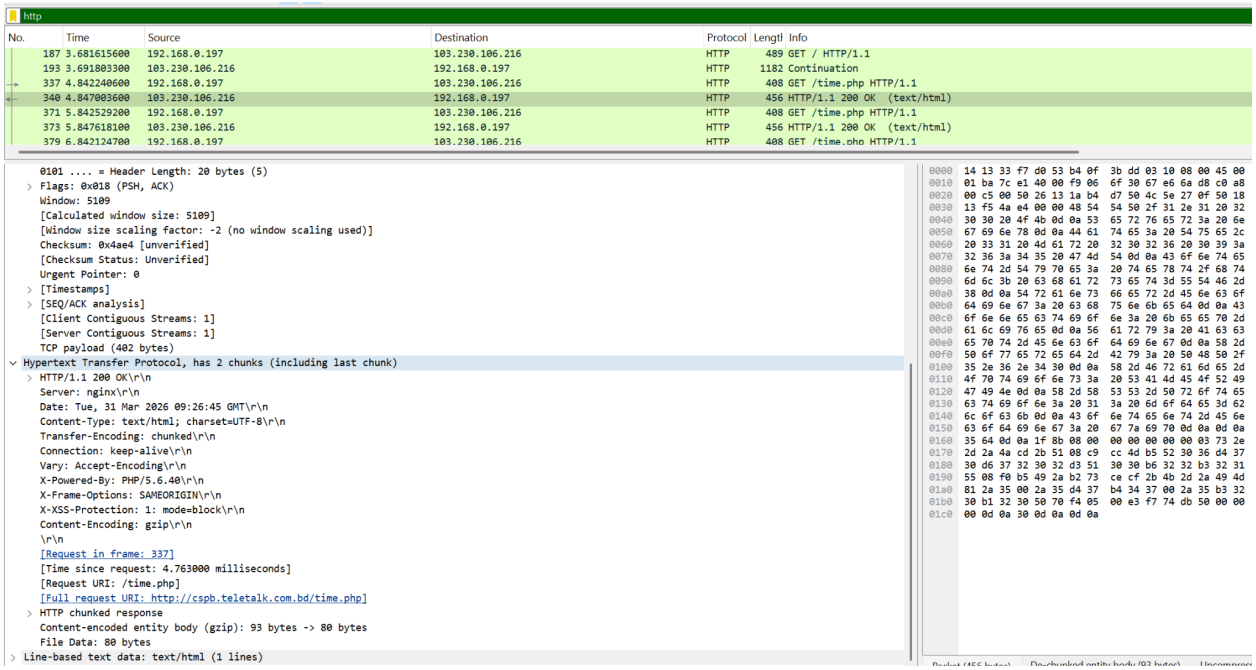
When a link is clicked, the computer bundles the request into a structured stack of data to ensure it travels safely across the network. At the Application Layer, the browser creates an HTTP GET request, which is the actual message asking the Teletalk server for its homepage. This request is handed to the Transport Layer (TCP), which adds the browser's return port (9747) and the server's destination port (80). These ports act like internal apartment numbers, ensuring that when the website responds, the data goes back to the correct browser tab rather than a different app.

The Network Layer (IPv4) then wraps this information in a global envelope by attaching the computer's local IP address (192.168.0.197) as the sender and the server's IP (103.230.106.216) as the receiver. This allows the packet to be routed across the internet to the correct machine. Finally, at the Data Link Layer (Ethernet), the computer uses MAC addresses to handle the first "hop" of the journey, marking the Wi-Fi adapter (14:13:33:f7:d0:53) as the source and the Tenda router (b4:0f:3b:dd:03:10) as the destination.

Once this entire package is ready, the Physical Layer converts the digital 1s and 0s into radio waves that travel through the air to the router. Each layer effectively adds its own header, MACs for local delivery, IPs for global routing, and Ports for application targeting, creating a 489-byte frame that tells the network exactly where to go and what to do.

HTTP Response Packet:





When the server responds, the process works like a return shipment being delivered. At the Application Layer, the server sends back an HTTP 200 OK message containing the requested data. In this case, a small, compressed timestamp from the server. This message is packed into the Transport Layer (TCP), which labels it with source port 80 (the server) and the destination port 9747. This ensures the data is directed specifically to the browser tab that made the request rather than any other open app on the computer.

The Network Layer (IPv4) then handles the global delivery by setting the source IP to the server (103.230.106.216) and the destination IP to the local PC (192.168.0.197). As the packet arrives at the network, the Data Link Layer takes over for the final "hop," using the Tenda router's MAC address (b4:0f:3b:dd:03:10) as the sender and the Wi-Fi adapter (14:13:33:f7:d0:53) as the receiver.

Finally, at the Physical Layer, these digital headers and data are transmitted as radio waves to the device. Once received, the computer goes through each layer, stripping away the MAC, IP, and Port information to reveal the underlying HTTP response.